BACTERIA AND ITS ECONOMIC IMPORTANCE

📝 Part 1: In-Chapter Point-Wise Notes 🦠🔬

🧬 9.1 Bacteria - A General Study

- **Nature:** Bacteria are the most primitive, unicellular organisms on Earth. They are strictly **prokaryotes**, meaning their genetic material is dispersed in the central region of the cytoplasm without a true nuclear membrane.
- **Classification Fact:** According to the Five Kingdom classification, bacteria are **not plants** because they are prokaryotic and single-celled, rather than eukaryotic and multicellular.
- **Occurrence & Benefit:** They are present everywhere (air, water, soil, food, and inside living bodies). Some bacteria residing in the human intestine are beneficial as they synthesize vitamins like Vitamin B, while bacteria living normally on our skin prevent the growth of harmful pathogens.
- **Size:** They are the smallest living organisms. Their average size is 2 micrometres long and 0.5 micrometre thick (where 1 micrometre = 1/1000th of a millimetre).

📐 Four Common Shapes & Arrangements of Bacteria

- **(1) Cocci:** Spherical bacteria.
- **(2) Bacilli:** Rod-shaped bacteria.
- **(3) Spirilla:** Spiral or twisted bacteria.
- **(4) Vibrio:** Comma-shaped bacteria.
- **Arrangement Patterns:** They can live singly, in pairs (**diplococci**), in long chains (**streptococci**), or in bunches/clusters (**staphylococci**).

🧱 Structure of a Bacterial Cell

- **Cell Wall:** Contains a non-living, stiff cell wall made of **peptidoglycan**, strictly *not* of cellulose as found in plant cells.

- **Cell Membrane & Cytoplasm:** A thin cell membrane lies directly beneath the wall, surrounding the protoplasm. The cytoplasm contains vacuoles, glycogen granules, and occasionally chlorophyll, but lacks a well-defined nucleus.
- **Genetic Material:** Chromatin (DNA) is present as a **single circle** (bacterial chromosome) located in the central region, attached at a point to the cell membrane.
- **Plasmid:** An extra, small circular DNA molecule inside the cell that carries **non-essential genes**.
- **Capsule & Flagella:** Sometimes a slimy, protective layer called a **capsule** coats the outside of the cell wall. While most bacteria are passively transported, some possess whip-like **flagella** that pierce through the capsule to provide active locomotion in liquids.

🍽️ Nutrition & Respiration

- **Nutrition:** Most are heterotrophic (lacking chlorophyll).
    - **Saprotrophic:** Draw nourishment from decaying dead organisms.
    - **Parasitic:** Draw nourishment from living hosts.
    - *Mechanism:* They secrete powerful enzymes externally into their food source to make it soluble, then absorb the solution.
- **Respiration:** Can be **aerobic** (absorbing atmospheric oxygen) or **anaerobic** (requiring no free oxygen; these are actually killed if exposed to air).

🔄 Reproduction & Survival

- **Asexual Reproduction (Fission):** The primary method. The circular DNA duplicates, the cell expands and constricts at the middle, separating into two identical daughter cells. Under ideal conditions, they divide incredibly fast (once every half hour).
- **Sexual Reproduction (Conjugation):** A highly primitive process where two bacteria of the same species attach. The **plasmid** (donor DNA) is transferred to the recipient through a hollow, tube-like extension.

- **Spore Formation:** Used purely to overcome unfavourable conditions (like extreme dryness), **not** as a method of reproduction.
    - The bacterial contents draw into a spherical mass covered by a thick, hard protective wall.
    - These spores are virtually indestructible; they tolerate extreme dryness, boiling water, frozen ice, and poisonous chemicals. They germinate back into active bacteria only when conditions become favourable again.

💊 9.2 Useful Role of Bacteria in Medicine (Antibiotics)

- **Antibiotics Definition:** A chemical substance produced by a living microorganism which can stop the growth of, or completely kill, disease-producing bacteria and fungi.
- **Penicillin:** The very first antibiotic for medical use. Discovered in 1929 by Alexander Fleming from a mould called *Penicillium chrysogenum* (formerly *P. notatum*).
    - *Key Fact:* Penicillin is the product of a **mould (fungus) and not of a bacterium!**
- **Streptomycin:** Discovered next by Selman Waksman, and this one *was* sourced from a bacterium named *Streptomyces griseus*. Other common antibiotics include chloromycetin and tetracycline

💊 9.2 Useful Role of Bacteria in Medicine (Continued)

1. Antibiotics & Their Sources

- **What makes a good antibiotic?**
    1. It should be "broad spectrum" (able to kill a wide variety of disease-producing microorganisms).
    2. It should not produce undesirable side-effects.
    3. It should not kill the normal, helpful bacteria of the host.
- **Other uses:** Used as food preservatives (especially for fresh meat and fish), treating animal feed, and controlling plant pathogens.

- **Key Antibiotics to Memorize:**
    - **Penicillin:** Sourced from *Penicillium chrysogenum* (a **mould/fungus**). Treats Tetanus and Diphtheria.
    - **Streptomycin:** Sourced from *Streptomyces griseus* (a **bacterium**). Treats Tuberculosis.
    - **Chlorotetracycline:** Sourced from *Streptomyces aureofaciens* (a **bacterium**). Treats Typhoid.
    - **Erythromycin:** Sourced from *Saccharopolyspora erythraea* (a **bacterium**). Treats Rickettsial fevers.

2. Serums & Toxins

- **Serum Definition:** Blood plasma from which the clotting protein *fibrinogen* has been removed. It helps in clotting and contains numerous chemical substances including antibodies and antitoxins.
- **Toxins:** Poisonous proteins released by pathogenic bacteria either during their active growth or upon their breakdown after they die.
- **Antitoxins:** Substances naturally produced in animal bodies that react with and neutralize the poison (toxin) produced by invading germs.
- **How Serums are Prepared:** A small dose of bacterial toxin is injected into the blood of a healthy animal (like a horse or cow). The body produces antitoxins to neutralize it. Blood is taken, chilled, and clears into a straw-coloured liquid called serum. *(Example: Snake-bite treatment uses anti-venine from horse blood).*
- **Genetic Engineering in Serums:** Human genes can be introduced into bacteria like *E. coli*.
    - The hormone **insulin** was the very first such substance produced by *E. coli*.
    - Also produces Blood Clotting Factor VIII (used when absent for Haemophilia A) and Factor IX (used when absent for Haemophilia B).

3. Vaccines & Toxoids

- **Vaccine:** A preparation consisting of weakened or dead germ substances. When injected, the body gets a mild form of the disease and produces **antitoxins** for future immunity.
    - *Killed bacteria:* TAB vaccine for typhoid.
    - *Living weakened bacteria:* BCG vaccine for tuberculosis.
- **Toxoids:** Inactivated/weakened toxins from particular bacteria specifically put into a person to stimulate the production of antibodies (used for immunity against diphtheria and tetanus).

🌱 9.3 Bacteria - Role in Agriculture (The Nitrogen Cycle)

- **1. Nitrogen-Fixing Bacteria:** Special bacteria (like *Rhizobium*) live in small nodules on the roots of leguminous plants (e.g., beans). They pick up free nitrogen from the soil/atmosphere and convert it into soluble nitrates.
- **2. Free-Living Soil Bacteria:** Bacteria like *Azotobacter* and *Clostridium* live freely in soil. They use free nitrogen, convert it to ammonia (NH_3) with hydrogen, which then turns into amino acids and nitrates, enriching the soil.
- **3. Nitrifying Bacteria (Nitrification):** Plants absorb nitrates for proteins.
    - *Step 1:* Nitrogen wastes and dead plants/animals convert into ammonia.
    - *Step 2:* Ammonia is converted into **nitrites** by bacteria like *Nitrosomonas*.
    - *Step 3:* Nitrites are converted into **nitrates** by bacteria like *Nitrobacter*.
- **4. Denitrifying Bacteria:** Present in the soil (e.g., *Pseudomonas*), they break down soil nitrates to release free nitrogen gas (N_2) back into the atmosphere.

🏭 9.4 Bacteria - Role in Industry

- **Biogas (Gobar Gas):** Fermenting bacteria degrade cellulose in cow dung to release highly inflammable methane gas. This provides both fuel and rich manure.
- **Internal Digestion:** Bacteria in our intestines habitually synthesize Vitamin B complex and Vitamin K. In herbivores, they digest cellulose.
- **Manufacturing:** Used in producing vinegar, processing coffee/tobacco, and **retting** (loosening fibres of hemp/flax by rotting).
- **Specific Roles:**
    - **Tea Curing:** Different flavours of tea are produced by certain bacteria.
    - **Leather Tanning:** Brought about by sunlight and bacteria breaking down the soft, perishable parts of animal skin (hide).

🚫🍔 9.5 Spoilage of Food & Preservation Methods

- **Botulism:** A very serious form of food poisoning caused by a special bacterium found in tinned/sealed foods. Affected cans become distended with gas which gushes out on opening—these must be strictly discarded.
- **Key Food Preservation Methods:**
    1. **Boiling (Sterilization):** Kills all bacteria except spores. Higher temperatures (around 110°C) at increased pressure (like in an autoclave or pressure cooker) will kill even the tough spores.
    2. **Salting:** Mixing foods with salt prevents bio-degradation (called *curing*).
    3. **Dehydration (Drying):** Microbes cannot grow without water (e.g., dried milk powder).
    4. **Irradiation:** Ultra-violet light is used to sterilize air in hospitals/bakeries to completely kill and free the area of germs. Radioactive radiation is rarely used for foods.
    5. **Pasteurization:** Partial sterilization (usually for milk). Milk is heated to **60°C for 30 minutes** and chilled quickly. *Note:* It kills

the majority of bacteria to prevent quick souring, but it does **not** completely sterilize the milk.

6. **Refrigeration:** Microbes do not grow at or below the freezing point (0°C to -5°C). Deep freezing utilizes -20°C to -30°C.
7. **Strong Concentration & Chemicals:** Jams and pickles use high concentrations of sugar and salt. This hypertonic environment causes **plasmolysis** (the bacteria literally lose water and die). Chemical preservatives like sodium benzoate are also used in tinned foods.

😳 9.6 Bacterial Diseases & Bioweapons

- **A. In Plants:**
  - *Black rot* of mustard and cauliflower.
  - *Bacterial blight* of cowpea (lobia).
- **B. In Animals (Cattle):**
  - *Anthrax* (causes swelling on the body and reduced milk yield).
  - *Bovine tuberculosis* (lungs affected, causing a dry husky cough).
- **C. In Humans:**
  - Whooping cough, cholera, tuberculosis, diphtheria, typhoid, pneumonia, and tetanus.
- **9.7 Bioweapons:** The terrifying act of intentionally sending or releasing bacteria as weapons. Some countries prepare 'germ bombs' which, when exploded, release epidemic-causing germs such as **Anthrax** bacteria to cause widespread death generation after generation.

QUESTION AND ANSWERS

📘 A. MULTIPLE CHOICE TYPE

- **(i) Bacteria are no more classified as plants primarily because:**
  - **(a) they are unicellular and prokaryotic.**

- **(ii) A particular species of which one of the following, is the source bacterium of the antibiotic, discovered next to penicillin, for the treatment of tuberculosis?**
    - **(b) *Streptomyces*** *(Specifically, Streptomyces griseus for streptomycin).*
- **(iii) Which bacteria is rod shaped?**
    - **(c) Bacillus.**
- **(iv) Which bacteria fixes nitrogen in the soil?**
    - **(c) *Rhizobium*** *(Note: (d) Clostridium is also technically correct as a free-living soil fixer, but Rhizobium is the classic primary answer for this level!).*
- **(v) Cell wall of bacteria is composed of:**
    - **(b) Peptidoglycan.**
- **(vi) Which of the following structures is suspended in the cytoplasm of a bacterial cell?**
    - **(c) Plasmid.**
- **(vii) Bacteria mainly reproduces by means of:**
    - **(c) Binary fission.**
- **(viii) Bacteria are refered to as prokaryotes because:**
    - **(d) They do not have a well defined nucleus enclosed in a nuclear membrane..**
- **(ix) Which of the following diseases can be prevented by TAB?**
    - **(b) Typhoid.**
- **(x) A preparation consisting of weakened or dead germ substances is:**
    - **(d) Vaccines.**
- **(xi) Certain free-living bacteria in the soil like (P) ..... and (Q) ..... use the free nitrogen of the soil and convert it to ammonia with hydrogen. P and Q are:**

- **(b) P — *Azotobacter*, Q — *Clostridium*.**

📕 B. VERY SHORT ANSWER TYPE

1. Name the following:

- **(a) The locomotory structure of a bacterium that enables it to swim:** Flagellum.
- **(b) The process of flavouring tea by certain bacteria:** Tea curing.
- **(c) The process of making leather from the animals' skin:** Leather tanning.
- **(d) Bacterial disease of mustard:** Black rot.
- **(e) The scientist who discovered the antibiotic streptomycin:** Selman Waksman.

2. Match the items in Column A with those in Column B: 🧩

- **(i) *Penicillium* ➡️ (b) Antibiotic.**
- **(ii) Diplococci ➡️ (c) Bacteria occurring in pairs.**
- **(iii) Streptococci ➡️ (a) Bacteria occurring in chains.**

3. Fill in the blanks: ✍️

- **(a)** The first antibiotic penicillin was discovered by **Alexander Fleming**.
- **(b) *Rhizobium*** bacteria is found living in the nodules on the roots of leguminous plants.
- **(c)** The hormone **insulin** was the first substance produced by *Escherichia coli*.
- **(d) *Pseudomonas*** is an example of denitrifying bacteria.
- **(e) Saprotrophic** and **parasitic** are the two heterotrophic modes of nutrition in bacteria.

4. Given below is the structure of a bacterial cell. Match the structures marked (a) to (e) with their function or characteristics given: 🔬

- **(a) Flagellum ➡️ 2. Locomotory structure..**

- **(b) Bacterial Chromosome (DNA) ➡️ 3. Composed of DNA..**
- **(c) Glycogen granules ➡️ 5. Stored food material..**
- **(d) Cell Wall ➡️ 4. Made up of peptidoglycan..**
- **(e) Capsule ➡️ 1. Protective layer..**

📘 **B. VERY SHORT ANSWER TYPE (Continued from Page 106)**

- **5. Name the bacteria that converts:**
  - **(a) ammonium compounds to nitrites:** *Nitrosomonas*.
  - **(b) nitrites to nitrates:** *Nitrobacter*.
- **6. Flow chart of 'Nitrogen cycle' blanks:**
  - (a) **nitrification**.
  - (b) **ammonia**.
  - (c) **ammonium**.
  - (d) ***Nitrosomonas***.
  - (e) ***Nitrobacter***.
  - *(Note on the diagram: P = Ammonia/Ammonium, Q = Nitrites, R = Nitrates).*

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- **5. Give two examples of each:**
  - **(a) Bacterial diseases of cattle:** Anthrax, Bovine tuberculosis.
  - **(b) Bacterial diseases in plants:** Black rot of mustard, Bacterial blight of cowpea.
  - **(c) Bacterial diseases in humans:** Cholera, Tuberculosis (or Typhoid, Diphtheria).
  - **(d) Nitrifying bacteria:** *Nitrosomonas*, *Nitrobacter*.
  - **(e) Free living nitrogen-fixing bacteria found in soil:** *Azotobacter*, *Clostridium*.
- **6. Write the full forms of the following abbreviations:**

- **(a) TAB:** Typhoid-Paratyphoid A and B vaccine.
- **(b) BCG:** Bacillus Calmette-Guérin.
- **(c) DNA:** Deoxyribonucleic acid.
- **(d) GMO:** Genetically Modified Organism.

- **7. Give reasons:**
  - **(a) Sunlight in rooms:** Direct sunlight contains ultra-violet rays which act as a natural sterilizing agent, killing harmful bacteria and mould spores floating in the air.
  - **(b) Tinned and sealed food safety:** It is not always safe because it can be infected by a special bacterium causing Botulism (a severe form of food poisoning), where the cans get distended with gas.
- **8. State the differences between:**
  - **(a) Decay and putrefaction:** Decay is the *complete* breakdown of organic matter without giving out a foul smell, whereas putrefaction is the *incomplete* breakdown of organic matter that emits a foul smell.
  - **(b) Pasteurization and sterilization:** Pasteurization is a technique of *partial* sterilization (heating to 60°C for 30 minutes) that kills a majority of bacteria but not all. Sterilization (like boiling at high temperatures and pressure) kills *all* bacteria, including their highly resistant spores.
- **9. Would there be any bacteria in an aquarium?**
  - Yes, absolutely. Bacteria are present in the water, gravel, and filters to break down the excretory wastes of the animals and any dead plant matter.

📕 C. SHORT ANSWER TYPE

- **1. In what form bacteria may be present in the air?**
  - They are primarily present in the form of highly resistant **spores** (a resting stage to withstand extreme conditions).

- **2. Why is spore formation in bacteria not considered a form of reproduction?**
    - Because it does not multiply the number of bacteria. One bacterial cell forms one spore to escape unfavourable conditions, and upon returning to favorable conditions, it germinates back into exactly one bacterium.
- **3. In what respect do you consider bacteria as simple organisms?**
    - They are extremely primitive, single-celled **prokaryotes**. They lack a well-defined nucleus enclosed in a nuclear membrane and do not possess complex cell organelles.
- **4. Define the terms:**
    - **(a) Antibiotics:** Chemical substances produced by living microorganisms which can stop the growth of or completely kill disease-producing bacteria and fungi.
    - **(b) Antitoxins:** Substances naturally produced in animal bodies which react with and neutralize the poison (toxins) produced by invading germs.
    - **(c) Serum:** Blood plasma from which fibrinogen has been removed, containing specific antitoxins used as a preventive measure against bacterial invasion.
    - **(d) Botulism:** A very serious form of food poisoning caused by a special bacterium sometimes found in improperly tinned and sealed foods.
    - **(e) Pasteurization:** A technique of partial sterilization applied to milk by heating it to about 60°C for 30 minutes and then chilling it quickly.

📗 D. LONG ANSWER TYPE

- **1. How do bacteria obtain their nourishment?**
    - Most are **heterotrophic**. They obtain food either as *saprotrophs* (drawing nourishment from decaying dead organisms) or as *parasites* (drawing nourishment from living hosts). They secrete

powerful enzymes outside their cell to dissolve the food material, which is then absorbed as a solution into the bacterial cell.

- **2. Describe any two uses of bacteria in industry.**
    - **(1) Leather Tanning:** Sunlight and certain bacteria are used to break down the soft, perishable parts of animal skin (hide) to create leather.
    - **(2) Tea Curing:** Certain bacteria are utilized to produce and enhance the different specific flavours of tea.
- **3. What are antibiotics? Name any two examples.**
    - They are chemical substances produced by a living microorganism which can stop the growth of or kill disease-producing microbes. Examples: **Penicillin** and **Streptomycin**.
- **4. Briefly describe how nitrogen of the atmosphere is converted to nitrates by leguminous plants.**
    - Symbiotic bacteria (like *Rhizobium*) live inside small root nodules on leguminous plants. They pick up free nitrogen directly from the soil and atmosphere and convert it into soluble nitrates, which are then absorbed and utilized by the host plant.
- **5. State how the nitrates in the soil get converted to nitrogen of the atmosphere.**
    - This is carried out by **denitrifying bacteria** (like *Pseudomonas*) present in the soil. They break down the soil nitrates to release pure nitrogen gas (N_2) that directly enters back into the atmosphere.

🎨 E. STRUCTURED / APPLICATION / SKILL TYPE

- **1. (a) Define the terms:**
    - **(i) Nitrification:** The biological process by which some bacteria first convert nitrogenous wastes/ammonia into nitrites (via *Nitrosomonas*), and then into nitrates (via *Nitrobacter*).

    - **(ii) Denitrification:** The biological breakdown of soil nitrates by specific soil bacteria to release free nitrogen gas back into the atmosphere.

- **1. (b) Distinguish between biological nitrogen fixation and industrial nitrogen fixation:**
    - **Biological Fixation:** The natural process performed by living organisms (like *Rhizobium* or *Azotobacter*) that convert atmospheric nitrogen into usable nitrates.
    - **Industrial Fixation:** A human-made, chemical process carried out in factories to synthesize nitrogenous fertilizers from atmospheric nitrogen.
- **1. (c) Name the kind of relationship that exists between *Rhizobium* and leguminous plants:**
    - It is a **Symbiotic relationship** (Symbiosis).

🤔 F. ASSERTION - REASON TYPE QUESTIONS

- **(i)** Answer: **(a) Both A and R are true and R is the correct explanation of A.** *(Bacteria are prokaryotes because their single circular chromatin is not enclosed in a nuclear membrane).*
- **(ii)** Answer: **(d) A is false but R is true.** *(Assertion is false because saprotrophic means drawing nourishment from DEAD organisms, not living. The Reason statement defining parasites is true).*
- **(iii)** Answer: **(d) A is false but R is true.** *(Assertion is false because Clostridium is a FREE-LIVING soil bacteria, not found in root nodules. Rhizobium is in the nodules).*
- **(iv)** Answer: **(a) Both A and R are true and R is the correct explanation of A.** *(Pasteurization is specifically considered partial sterilization because it only kills a majority of bacteria, not all of them).*
- **(v)** Answer: **(d) A is false but R is true.** *(Assertion is false because both black rot of mustard and cauliflower are purely bacterial diseases, as confirmed by the Reason).*

🌎 G. CASE STUDY BASED QUESTION (Nazia's Farm)

- **(a) Write the scientific name of the organism found in these swollen parts of the root.**
    - ***Rhizobium***.
- **(b) What term is used to describe these swollen structures of the root?**
    - **Root nodules**.
- **(c) Mention the category of the organisms, answered in (b) above.**
    - **Nitrogen-fixing bacteria** (or Symbiotic bacteria).
- **(d) How do these organisms improve the fertility of the soil? Name the process.**
    - They pick up free nitrogen from the soil and atmosphere and convert it into soluble nitrates, thus enriching the soil. The process is called **Biological Nitrogen Fixation**.
- **(e) Name two compounds which are produced during the above mentioned process.**
    - **Nitrates** and **Amino acids**.

🐟 **BLUE BOX: WOULD THERE BE ANY BACTERIA IN AN AQUARIUM?**

- **Name any two such animals:** Fishes and Snails.
- **Do you think there would be some bacteria in it? If so, in which part... and what for?**
    - **Yes**. Bacteria would occur in abundance in the **soil/gravel at the bottom** and inside the water.
    - **What for:** They are essential for decaying and breaking down the organic excretory wastes produced by the fishes and any dead plant matter, recycling nutrients back into the mini-ecosystem.